Supplementary Information

Integrated Optimization of Urban Morphology and Energy Performance through Deep Reinforcement Learning: A Block-Scale Application

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S-1: Case Study and Parametric Modeling Details

S-1.1 Block Geometric Parameters and Modeling Logic

The parametric modeling for this study was developed on the Grasshopper platform. The selected study area is Jianhu City, located in the eastern coastal area of China 33° 16'to33°41'N, 119°33' to 120°05' E). This area was chosen as a case study based on its unique urban fabric and climate context. First, Jianhu's urban development provides a rich and diverse sample of residential building types, ranging from 3-story low-rise buildings to 30-story high-rise buildings, offering a solid foundation for constructing representative prototypes. Second, the region's climate profile presents a significant challenge for building energy consumption. According to the Chinese "Thermal design code for civil building," Jianhu is located in the "hot summer and cold winter" zone and is classified as a Cfa climate (humid subtropical climate) in the Köppen-Geiger classification. The city's mean annual dry bulb temperature is 15°, with the hottest month (July) averaging 27° and the coldest month (January) averaging 1。6°. This climate profile necessitates significant energy demand for both building heating in winter and cooling in summer, making it an ideal location for energy performance optimization studies of building and urban form. The annual solar radiation in this region is approximately 1250 kWh/m^2.

To ensure the representativeness of the study, we first analyzed the existing urban fabric of Jianhu to determine a typical and dominant block scale. The analysis indicated that a 240 m × 240 m dimension best represents the prevailing local street grid layout.

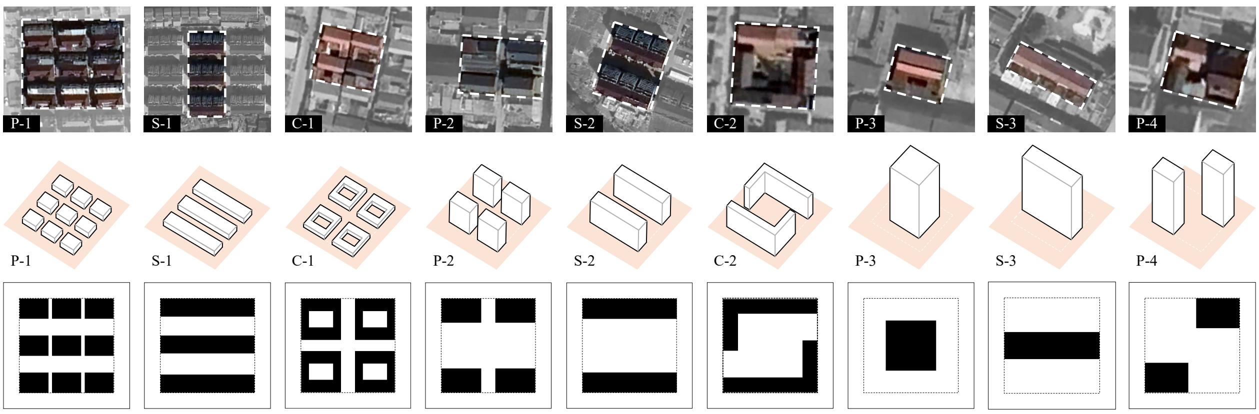

Fig. S1. Selection of building types from the urban context in Jianhu city

Therefore, the base modeling unit for this study was defined as a hypothetical block of 240 m × 240 m.

As shown in Fig. S1, this block is systematically subdivided into a 3×3 grid, creating nine uniform 80 m × 80 m land units (numbered 0–8). To simulate realistic urban planning constraints and define the buildable area, we applied a precise geometric modeling logic:

(1) Road Boundary: The boundary line of each 80 m × 80 m land unit is first offset by 5 m to the inside. This line is defined as the road boundary.

(2) Building Boundary (Setback): This newly generated road boundary is then offset by another 10 m to the inside. This second line defines the building boundary, enforcing a setback.

(3) Buildable Area: This two-step offset process results in a final buildable area of 60 m × 60 m within each land unit.

In the subsequent optimization process, one of the nine land units is designated as open space, while the remaining eight units can each accommodate one building type. To ensure simulation consistency, the floor-to-floor height for all buildings is uniformly set to 3 m. This parametric modeling framework allows the model to generate a vast array of distinct urban block forms by varying the building type, building height, open space location, and block orientation.

To ensure the generated forms are relevant to the local context, we constructed and abstracted nine building prototypes based on a detailed analysis of Jianhu's urban fabric, as shown in Fig. S2. These nine prototypes are grouped into three main categories based on their plan geometry:

- Point Buildings (P-1 to P-4): Typically characterized by a smaller, more compact footprint.
- Slab Buildings (S-1 to S-3): Characterized by an elongated, rectangular footprint.
- Courtyard Buildings (C-1 and C-2): Characterized by a building mass enclosing a central open space.

These prototypes also represent the full spectrum of development densities found in Jianhu's residential development:

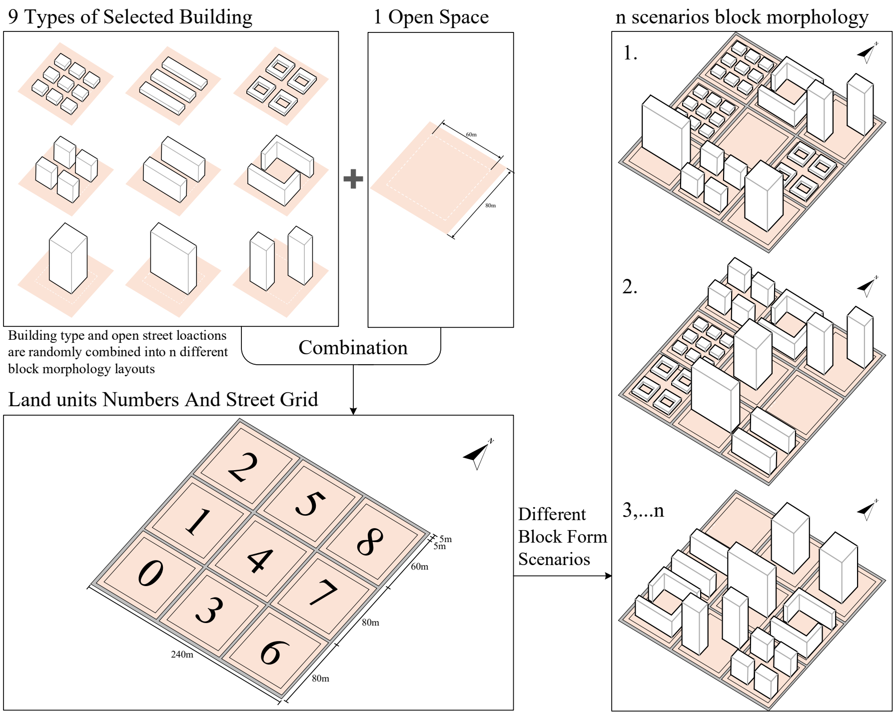

Fig. S2. Model block form in Rhino

- P-1, S-1, and C-1 correspond to low-density, low-rise forms (e.g., semi-detached houses, row/terraced houses, and low-rise courtyard houses).
- P-2, S-2, and C-2 represent medium-density, multi-story collective residential buildings.
- P-3, P-4, and S-3 represent low-density, high-rise collective residential buildings.

This prototype selection is comprehensive, as they are the most typical forms found in Jianhu's residential development, collectively covering over 95% of the existing residential building forms. To simplify the model and enhance computational efficiency, while also simulating realistic development practices (where a limited number of building types are often chosen for cost control), this study standardized the building types. Building types within the same height variation range were simplified to share the same footprint area and building density. The height of each building is categorized into three distinct ranges: low-rise (1–3 floors), mid-rise (4–12 floors), and high-rise (13–30 floors). The specific morphological parameters, footprint areas, and constraints for each prototype are detailed in Table S1.

Finally, the parametric model described above allows for the precise calculation of a series of key Urban Form Factors (UFFs) for each generated block. These factors serve

TABLE SI

MORPHOLOGICAL FACTORS OF NINE BUILDING TYOES

ID	Building Type	Footprint Area	Number of Floors	Building Density	FAR Range
P-1	Point buildings	2000	1-3	31.3	0.31 - 0.94
P-2	Point buildings	1500	4-12	23.4	0.94 - 2.81
P-3	Point buildings	1000	13-30	15.6	2.03 - 4.69
P-4	Point buildings	1000	13-30	15.6	2.03 - 4.69
S-1	Slab buildings	2000	1-3	31.3	0.31 - 0.94
S-2	Slab buildings	1500	4-12	23.4	0.94 - 2.81
S-3	Slab buildings	1000	13-30	15.6	2.03 - 4.69
C-1	courtyard buildings	2000	1-3	31.3	0.31 - 0.94
C-2	courtyard buildings	1500	4-12	23.4	0.94 - 2.81

as the quantitative descriptors of the urban morphology. In the main body of this study, these factors are critical variables, serving not only as the State Representation for the deep reinforcement learning (DRL) agent but also as the basis for the subsequent performance correlation analysis. The primary factors extracted from each geometric model include:

Density Metrics: These quantify the development intensity of the block.

- Floor Area Ratio (FAR): The total gross floor area of all buildings divided by the total block site area.
- Building Density (BD): The total building footprint area divided by the total block site area.
- Open Space Rate (OSR): The non-built-up area of the block divided by the total gross floor area.
- Average number of Floors (AF): The total gross floor area divided by the total building footprint area.

Geometric and Compactness Metrics: These describe the geometric properties of the buildings themselves, which directly influence thermal performance.

- Shape Coefficient (SC): The total surface area of a building divided by its total gross floor area. This is a crucial indicator of compactness; a lower SC generally implies less external envelope area for heat loss or gain per unit of floor area.
- Average Perimeter-Area Ratio (PAR): The average ratio of a building's perimeter to its footprint area. Similar to SC, it reflects the building's ability to obtain solar radiation and dissipate heat.

Spatial Configuration Metrics: These describe the 3D arrangement and openness of the block.

- Mean Sky View Factor (SVF): This factor quantifies the fraction of the sky hemisphere visible from a point on the ground. A higher SVF indicates a more open area, which has a significant positive correlation with rooftop PV energy generation as it implies less shading from surrounding buildings.

S-2: Surrogate Model Validation with K-Fold Cross-Validation

To rigorously assess the predictive performance and generalization capability of the DNN surrogate model, a 5-fold cross-validation (K=5) was conducted. The complete dataset (N=500) was randomly shuffled and partitioned into 5 unique, non-overlapping folds of 100 samples each.

The model was trained and evaluated 5 times. In each iteration k (from 1 to 5), the k-th fold was held out as the test set, while the remaining 4 folds (400 samples) were used as the training set. This ensures that the performance is evaluated across the entire dataset, mitigating potential bias from a single arbitrary split.

The detailed performance results (Coefficient of Determination, R^2, and Mean Absolute Error, MAE) for each of the 5 folds are presented in Table S1. The mean and standard deviation of these results are reported in the main manuscript (Table V) and confirm the high stability and accuracy of the surrogate model.

Objective	**Metric**	**Fold 1**	**Fold 2**	**Fold 3**	**Fold 4**	**Fold 5**	**Mean**	**Std. Dev.**
EUIt	R^2	0.97	0.985	0.975	0.962	0.973	0.973	0.008
	MAE	0.038	0.025	0.03	0.041	0.021	0.031	0.008
EG	R^2	0.96	0.978	0.982	0.955	0.97	0.969	0.011
	MAE	0.045	0.03	0.039	0.048	0.038	0.04	0.007
H	R^2	0.965	0.982	0.97	0.966	0.973	0.971	0.006
	MAE	0.033	0.02	0.029	0.03	0.027	0.028	0.005

S-3: Background of DDPG Algorithm

The Deep Deterministic Policy Gradient (DDPG) is a model-free, off-policy actor-critic algorithm designed to handle continuous action spaces.

The DDPG algorithm introduces a deterministic policy gradient for the actor, which updates the policy network parameters according to Equation (1).

$$\nabla_\theta J(\pi_\theta) = \int_s \rho^\pi(s) \nabla_\theta \pi_\theta(s) \nabla_a Q^\pi(s,a)\Big|_{a=\pi_\theta(s)} ds \tag{1}$$

where $\nabla_\theta J(\pi_\theta)$ is the policy gradient, $\rho^\pi(s)$ is the state sampling distribution, $\pi_\theta(s)$ is the deterministic action output by the policy network, and $Q^\pi(s,a)$ represents the Q-value estimated by the critic network.

The critic network updates its parameters by minimizing the Mean Squared Error (MSE), as shown in Equation (2):

$$L(\theta^Q) = E_{(s_t,a_t,r_t)}\left[\left(Q\left(s_t,a_t\middle|\theta^Q\right)-y_t\right)^2\right] \tag{2}$$

where y_t is the target Q-value calculated using the Temporal-Difference target, as shown in Equation (3):

$$y_t = r(s_t,a_t) + \gamma Q\left(s_{t+1}, \pi(s_{t+1}) \mid \theta^Q\right) \tag{3}$$

During training, however, directly using the current critic and actor networks to calculate the target value causes y_t to change rapidly with frequent parameter updates, which can lead to considerable fluctuations or even divergence in the learning process. To address this, DDPG adapts the target network mechanism from DQN by employing a smoother soft update strategy for the target network parameters. As shown in Equation (4), this gradual update method allows the target network parameters to slowly track the changes in the main network parameters, thereby improving the stability of the learning process. The introduction of the Experience Replay mechanism from DQN further improves the efficiency of parameter updates.

$$\theta^{\pi'} \leftarrow \rho\theta^\pi + (1-\rho)\theta^{\pi'}, \theta^{Q'} \leftarrow \rho\theta^Q + (1-\rho)\theta^{Q'}, \rho \ll 1 \tag{4}$$

where ρ is the soft update factor.

To enhance exploration, DDPG introduces random noise, enabling the agent to search more thoroughly for the optimal solution within the continuous action space. As defined in Equation (5), the addition of a random noise term, N, introduces stochasticity into the action selection, thereby achieving a degree of policy exploration.

$$\pi'(s_t) = \pi(s_t | \ \theta_t^{\pi}) + N_t \tag{5}$$

In summary, DDPG merges the advantages of value estimation and policy optimization to efficiently learn an optimal policy. It is therefore well-suited for optimization tasks involving block morphological factors, providing useful technical support and a theoretical reference for energy-efficient urban design and sustainable development.

Algorithm 1: DDPG

1: Initialize critic network $Q(s,a|\theta^Q)$ and actor network $\mu(s|\theta^\mu)$ with weights θ^Q and θ^μ
2: Initialize target networks Q' and μ' with weights $\theta^{Q'}$ and $\theta^{\mu'}$
3: Initialize a replay buffer R
4: **For** each episode e **do**
5: Initialize a random process u for action exploration
6: Receive the initial observation state s_1
7:   **For** each time step t **do**
8:   Select an action $a_t = \mu(s_t|\theta^\mu) + u_t$ according to the current policy and exploration noise
9:   Execute the action a_t and observe the reward r_t and the new state s_{t+1}
10:   Store the transition (s_t, a_t, r_t, s_{t+1}) in the replay buffer R
11:   Sample a random minibatch of N transitions (s_t, a_t, r_t, s_{t+1}) from the replay buffer
12:   Compute the target Q-value: $y_i = r_i + \gamma Q'\left(s_{i+1}, \mu'\left(s_{i+1} | \ \theta^{\mu'}\right) | \ \theta^{Q'}\right)$
13:   Update the critic by minimizing the loss: $L = 1/N \sum_i \left(y_i - Q\left(s_{i+1}, a_i | \ \theta^Q\right)\right)^2$
14   Update the critic by minimizing the loss:
$\nabla\theta^\mu J \approx 1/N \sum_i \nabla a Q\left(s, a \middle| \theta^Q\right)\Big|_{s=s_t, a=\mu(s_i)} \nabla\theta^\mu \mu\left(s \middle| \theta^\mu\right)\Big|_{si}$
15:   Update the target networks:
$\theta^{Q'} \leftarrow \tau\theta^Q + (1-\tau)\theta^{Q'}$
$\theta^{\mu'} \leftarrow \tau\theta^\mu + (1-\tau)\theta^{\mu'}$
16:   **End for**
17: **End for**

S-4: Detailed Statistical Comparison of Morphological Strategy Distributions

This section provides a detailed statistical visualization of the final solution clusters generated by the combined DDPG agents and the NSGA-II benchmark. These figures complement the analysis in the main manuscript (e.g., Fig. 8, Fig. 9) by offering a deeper insight into the distribution, central tendency, and variance of the 12 Urban Morphological Factors (UMFs) for each optimization method.

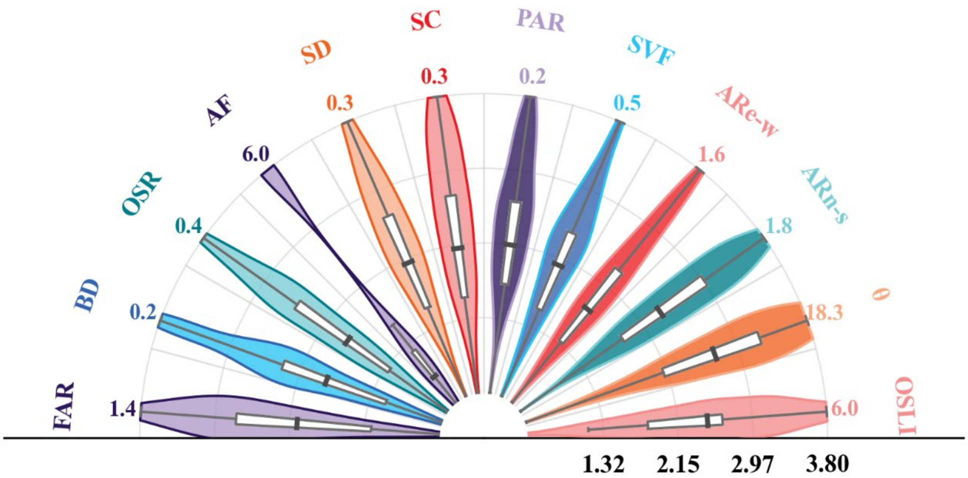

Fig. S3. NSGA-II solution set distribution across the 12 UMFs

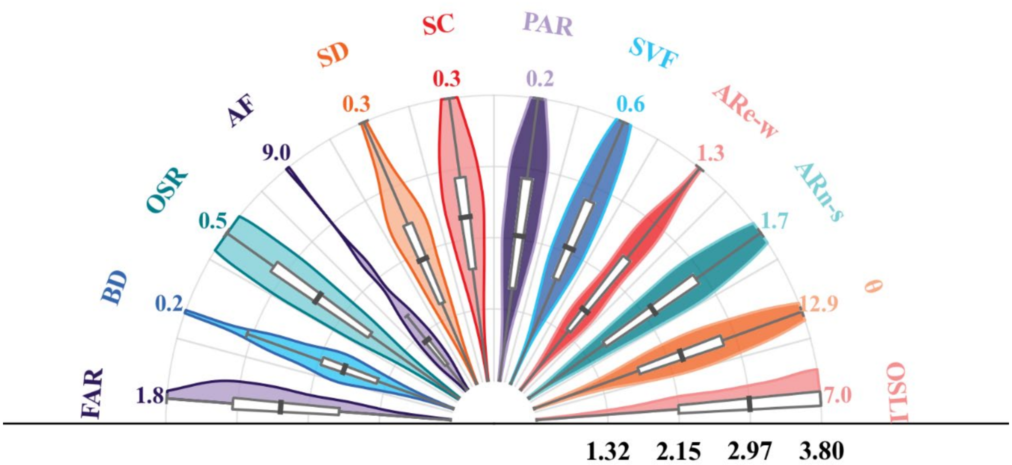

Fig. S4. Combined DDPG solution set distribution from all three scenarios

Both figures utilize semi-circular radial violin plots. The radius represents the variable's value, which is independently normalized within each variable's min-max range. The violin width indicates the sample density (Kernel Density Estimation). The inner white box represents the Interquartile Range (IQR, Q1-Q3), with the median (Q2) marked by a dark line and its numerical value labeled externally.

S-4.1 Plotting Methodology and Interpretation Guide

The visualizations in Fig. S4 are semi-circular radial violin plots, specifically designed to compare the statistical distributions of the 12 UMFs between the two main optimization approaches.

- **Data Source:** Fig. S4a displays the solution set generated by the NSGA-II algorithm (n≈133). Fig. S4b displays the aggregated solution set from all three DDPG specialist agents (n=150, 50 solutions from each) .
- **Axis Encoding (Critical Note):** It is essential to understand that the radial axis is **independently normalized for each of the 12 variables**. The values for each UMF (e.g., FAR, BD, OSR) are scaled to their *own* minimum-maximum range and then mapped to the uniform radial domain. **Therefore, the radial position and median value labels (e.g., 1.8 for FAR vs 0.5 for OSR) cannot be compared across different variables.** The utility of this plot lies in comparing the *distributional shape, median, and variance* of the *same variable* between the two methods (e.g., comparing the FAR plot in Fig. S4a to the FAR plot in Fig. S4b).
- **Statistical Representation:** For each variable, the plot shows:
    1. **Violin Plot (shaded area):** The width represents the Kernel Density Estimation (KDE) of the solutions. A wider "belly" indicates a higher concentration (density) of solutions at that specific value.
    2. **Box Plot (white box):** This represents the Interquartile Range (IQR), showing the range between the 25th percentile (Q1) and the 75th percentile (Q3).
    3. **Median (dark line):** This line marks the 50th percentile (Q2), with its precise numerical value labeled for clarity.

S-4.2 High-Level Observational Analysis

These plots visually substantiate key findings discussed in the main manuscript regarding the superiority and strategic focus of the DRL framework.

1. **DRL Convergence vs. NSGA-II Variance:** A primary observation is the

difference in distributional tightness. The DDPG solution set (Fig. S4b) generally exhibits more "pinched" or "tighter" distributions and more compact Interquartile Ranges (shorter white boxes) compared to the NSGA-II set (Fig. S4a). This is particularly evident in variables such as OSR, SD, SVF, and OSLI. This visualizes the DRL agents' strong convergence onto specific and stable design strategies, which is a core finding discussed in the main paper. In contrast, the NSGA-II plots often show wider, more spread-out distributions and taller IQR boxes (e.g., FAR, OSR, SVF), reflecting the high variance and "unfocused" nature of its solution set, which failed to converge on a single dominant strategy.

2. **Strategic Preference in Key Variables:** The plots clearly show the different morphological "fingerprints" learned by the two methods. For instance, the combined DRL agents (Fig. S4b) converged on a higher median OSR (0.5) compared to NSGA-II's median of 0.4. Furthermore, a stark difference is seen in the block orientation (θ), where the DRL solutions are highly concentrated around a median of 18.30, whereas the NSGA-II solutions are scattered around a much lower median of 8.35. This supports the main paper's conclusion that the DRL agents learned a distinct and strategically superior morphology that the benchmark algorithm failed to identify.